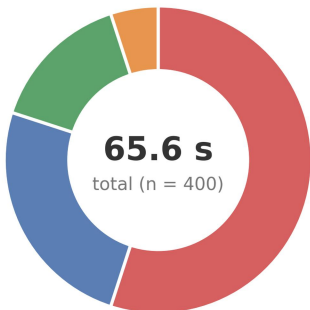


Conversion Time Breakdown by Phase (n = 400 rules, total 65.6 s)



	Projection Normalization	$O(N^2)$	55% (~36.1 s)
	Sweep-Line Decomp	$O(N \times n)$	25% (~16.4 s)
	Anomaly Detection	$O(n^2)$	15% (~9.8 s)
	Directional Merge	$O(N \log N)$	5% (~3.3 s)

$N = O(n^3)$ cells before merge (worst-case cut points)

Projection Normalization dominates at large n due to $O(N^2)$

Incremental update algorithms could reduce this (future work)

Conversion is a one-time offline cost — TRF deployed indefinitely